

torch.jit.isinstance#

<https://pytorch.org/docs/stable/generated/torch.jit.isinstance.html#torch.jit.isinstance>

Description:

Provide container type refinement in TorchScript. It can refine parameterized containers of the List, Dict, Tuple, and Optional types. E.g. List[str], Dict[str, List[torch.Tensor]], Optional[Tuple[int,str,int]]. It can also refine basic types such as bools and ints that are available in TorchScript. obj – object to refine the type of target_type – type to try to refine obj to

Function Signature

```
torch.jit.isinstance(obj, target_type)[source]#
```

Parameters

Returns

True if obj was successfully refined to the type of target_type,

Return type

Parameters

True if `obj` was successfully refined to the type of `target_type`,

False otherwise with no new type refinement

Returns:

bool

Code Examples

```
import torch
from typing import Any, Dict, List

class MyModule(torch.nn.Module):
    def __init__(self) -> None:
        super().__init__()

    def forward(self, input: Any): # note the Any type
        if torch.jit.isinstance(input, List[torch.Tensor]):
            for t in input:
                y = t.clamp(0, 0.5)
        elif torch.jit.isinstance(input, Dict[str, str]):
            for val in input.values():
                print(val)

m = torch.jit.script(MyModule())
x = [torch.rand(3,3), torch.rand(4,3)]
m(x)
y = {"key1": "val1", "key2": "val2"}
m(y)
```

Detailed Documentation

Documentation

Provide container type refinement in TorchScript.

It can refine parameterized containers of the List, Dict, Tuple, and Optional types. E.g. `List[str]`, `Dict[str, List[torch.Tensor]]`, `Optional[Tuple[int,str,int]]`. It can also refine basic types such as bools and ints that are available in TorchScript.

`obj` – object to refine the type of

`target_type` – type to try to refine `obj` to

False otherwise with no new type refinement

Example (using `torch.jit.isinstance` for type refinement): .. testcode: